

Exercise 2

Estimate solar neutrino flux at

- Earth
- Jupiter

Compare it to the available experimental flux.

Exercise 3

Consider two-neutrino oscillation with

- $E = 1 \text{ GeV}$

- $\sin^2 2\theta = 1$

- $\Delta m^2 = 2.5 \cdot 10^{-3} \text{ eV}^2$

Plot $P(\nu_\mu \rightarrow \nu_\tau)$ as a function of distance/(Earth diameter).